Ps 02

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1 Problem 1 DONE

1.1 Task

Let X be a metric space and $f : [0,1] \to X$ continuous. Show that $f([0,1]) \subseteq X$ is path-connected.

1.2 Solution

Lemma 11.1.12: Path-connectedness is preserved under continuous maps

Let X and Y be metric spaces. If X is path-connected and $f : X \to Y$ is continuous, then $f(X)$ is path connected.

Proof.

Let $a_1, a_2 \in f(X)$ be arbitrary. Then there are $x_1, x_2 \in X$ such that $f(x_1) = a_1$ and $f(x_2) = a_2$. Since X is connected, there is a continuous function $\gamma : [0,1] \to X$ such that $\gamma(0) = x_1$ and $\gamma(1) = x_2$. The function $f \circ \gamma : [0,1] \to f(X)$ is also continuous, because f is assumed to be continuous. It satisfies $f \circ \gamma(0) = f(x_1) = a_1$ and $f \circ \gamma(1) = f(x_2) = a_2$.

The interval $[0,1]$ is path-connected, since for any $s, t \in [0,1]$ the map $\gamma(u) = s + u(t - s)$ is a continuous path from s to t. Since f is continuous, $f([0,1])$ is path-connected by *Lemma 11.1.12*.

2 Problem 2 DONE

2.1 Task

Let $f : [0,1] \to \mathbb{R}^2$ and $f_1, f_2 : [0,1] \to \mathbb{R}$, so that

$$f(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \end{pmatrix}.$$

We equip $\mathbb{R}^2$ with the standard metric

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

(a) Prove that the coordinate map $\chi_i : \mathbb{R}^2 \to \mathbb{R}$ defined by $\chi_i(x_1, x_2) = x_i$ is continuous, for $i = 1, 2$.

(b) Prove that f is continuous if and only if both f_1 and f_2 are continuous.

2.2 Solution

2.2.1 (a)

Let $\epsilon > 0$. Choose $\delta = \epsilon$. Then if $d(x, y) < \delta$,

$$|\chi_i(x) - \chi_i(y)| = |x_i - y_i| \leq \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} = d(x, y) < \delta = \epsilon$$

Hence, $\chi_i : \mathbb{R}^2 \to \mathbb{R}$ is continuous.

2.2.2 (b)

- ($\Rightarrow$) If f is continuous, then $f_i = \chi_i \circ f$ is continuous as a composition of continuous maps.
- ($\Leftarrow$) Let $\epsilon > 0$. f_1 is continuous, so $\forall \epsilon > 0 \exists \delta_1 : |s - t| < \delta_1 \implies |f_1(s) - f_1(t)| < \frac{\epsilon}{\sqrt{2}}$. Similarly, for f_2, $\forall \epsilon > 0 \exists \delta_2 : |s - t| < \delta_2 \implies |f_2(s) - f_2(t)| < \frac{\epsilon}{\sqrt{2}}$.

  Define $\delta = \min(\delta_1, \delta_2)$. Thus,

$$d(f(s), f(t)) = \sqrt{(f_1(s) - f_1(t))^2 + (f_2(s) - f_2(t))^2} < \sqrt{\frac{\epsilon^2}{2} + \frac{\epsilon^2}{2}} = \epsilon$$

3 Problem 3 DONE

3.1 Task

Let $S = \mathbb{R}^2 \setminus \{(0, 0)\}$ and $x = (1, 0)$ and $y = (-1, 0)$.

(a) Define a continuous path from x to y.

(b) Plot this path in Julia.

(c) Visualize the sets

$$A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$
$$B = \{(-1, y) : -1 \leq y \leq 1\} \cup \{(1, y) : -1 \leq y \leq 1\}$$

in Julia by sample/grid points. Determine which of the two sets is path-connected.

3.2 Solution

3.2.1 (a)

$\gamma : [0,1] \to S \quad \gamma(t) = (\cos(\pi t), \sin(\pi t))$

3.2.2 (b)

```
using Plots

gamma(t) = (cos( * t), sin( * t))

ts = 0:0.01:1
xs = first.(gamma.(ts))
ys = last.(gamma.(ts))

plot(xs, ys,
     title="Continuous path",
     label="",
     xlabel="x", ylabel="y",
     aspect_ratio=:equal)
savefig("./assets/p_03_02.png")
```

```
/Users/ziro2173/personal/university/01_bachelor/02_semester/01_analysis/02_exercises/set_02/asse
```

3.2.3 (c)

```
using Plots

ys_range = -1:0.01:1

A_xs = [x for x in ys_range for y in ys_range if x^2 + y^2 < 1]
A_ys = [y for x in ys_range for y in ys_range if x^2 + y^2 < 1]

B_xs = [fill(-1, length(ys_range)); fill(1, length(ys_range))]
B_ys = [collect(ys_range); collect(ys_range)]

opts = (xlabel="x", ylabel="y", markersize=1, aspect_ratio=:equal, legend=false)

p_1 = scatter(A_xs, A_ys; title="Set A", opts...)
p_2 = scatter(B_xs, B_ys; title="Set B", opts...)

plot(p_1, p_2)
savefig("./assets/p_03_03_01.png")
```

Hence,

A is path-connected (straight line between any two points stays inside).

B is not path-connected (the two segments are disjoint).

4 Problem 4 DONE

4.1 Task

Let $f : [0,1] \to \mathbb{R}$ be given by

$$f(x) = \frac{1}{4}(x^2 + 1)$$

(a) Show that f maps $[0,1]$ into itself and is a contraction on $[0,1]$.

(b) Deduce that f has a unique fixed point and implement the fixed point iteration in Julia.

4.2 Solution

4.2.1 (a)

- f maps $[0,1]$ into itself.

$$\begin{aligned} x \in [0,1] &\implies x^2 \in [0,1] \\ &\implies x^2 + 1 \in [1,2] \\ &\implies \frac{1}{4}(x^2+1) \in \left[\frac{1}{4}, \frac{1}{2}\right] \end{aligned}$$

  As $\left[\frac{1}{4}, \frac{1}{2}\right] \subseteq [0,1]$, f maps $[0,1]$ into itself.

- f is a contraction on $[0,1]$. For all $x, y \in [0,1]$,

$$\begin{aligned} |f(x) - f(y)| &= \frac{1}{4}\left|x^2 - y^2\right| \\ &= \frac{1}{4}|x+y|\,|x-y| \\ &\leq \frac{1}{4} \cdot 2 \cdot |x-y| \\ &\leq \frac{1}{2}|x-y|\,. \end{aligned}$$

Hence f is a contraction with $q = \frac{1}{2}$.

4.2.2 (b)

```
f(x) = 1/4 * (x^2 + 1)

function fixed_point_iteration(f, x_0, tol=1e-10, max_iter=100)
    x = x_0
    for n in 1:max_iter
        x_new = f(x)
        if abs(x_new - x) < tol
            println("Converged after $n iterations")
            return x_new
        end
        x = x_new
    end
    @warn "Did not converge within $max_iter iterations"
    return x
end

x_star = fixed_point_iteration(f, 0.0)
println("Fixed point: $x_star")
println("Exact value: $(2 - sqrt(3))")
```

```
f (generic function with 1 method)
fixed_point_iteration (generic function with 3 methods)
Converged after 12 iterations
0.26794919242681425
Fixed point: 0.26794919242681425
Exact value: 0.2679491924311228
```